IBM FileNet P8 V5.5.8

Day 7

Workflow Development using Process Designer

Process Design

Routing Logic

Deploy & Test

Process Designer | Workflow Design | Build & Deploy

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Learning Objectives – Day 7

PD

Describe IBM Process Designer V5.5.8 role and interface components

PD

Design Process Maps with steps, connectors, and data fields

Logic

Implement routing logic: conditional branches and routing expressions

Logic

Configure workflow components: activities, sub-processes, escalations, timers

Build

Build 11-step DocumentApprovalWorkflow with rejection loop

Build

Configure Queue Steps with deadlines and escalation rules

Deploy

Validate workflow map and deploy .pep file to FileNet PE server

Deploy

Test end-to-end: launch, review, approve, and verify completion

01

Process Designer Introduction

Visual Workflow Design · .pep Files · PE Connection · Interface

IBM Process Designer V5.5.8 – Interface & Setup

Step Palette (Left)

Drag-and-drop step types: Launch, Queue, System, Notify, Map, Milestone, Abort, Terminate. All workflow building blocks.

Process Map Canvas (Center)

Main design area. Drop steps from palette, draw connectors between steps. Visual representation of the workflow.

Properties Panel (Right)

Properties of selected step or connector: queue name, routing expression, deadline, instructions, required fields.

Data Fields Panel (Bottom)

Define workflow variables (String, Integer, Date, Boolean). Name, initial value, visibility at launch.

Toolbar

Save (Ctrl+S), Open, Deploy to PE, Validate Map, Zoom. Key actions always accessible.

Map Menu

Map > Validate Map, Map > Deploy to Process Engine, Map > Properties. Deployment controls.

02

Workflow Design Components

Process Maps · Steps · Routing Logic · Conditional Branches

Workflow Step Types

Launch Step

Entry point. Exactly one per map. Defines data fields required at launch.

Queue Step

Assigns work item to a queue for human action. Most common step type.

System Step

Auto-executes. Contains JavaScript, component steps, or map steps.

Routing Decision

Diamond. Evaluates Boolean expression. Routes to different branches.

Notify Step

Sends email via PE Mailer. No queue assignment needed.

Map Step

Launches a sub-workflow. Can wait synchronously for child completion.

Milestone Step

Sets a named progress milestone. Used for SLA tracking and reporting.

Abort Step

Terminates the workflow immediately. Status set to TERMINATED.

Terminate Step

Circle. End of workflow. Every branch must reach Terminate.

Routing Logic & Conditional Branches

Connector Types

Unconditional	Always followed. No expression required. For mandatory sequential flow.
Conditional	Followed only when Boolean expression = TRUE. Example: F_Status == 'Approved'
Default	Dashed arrow. Followed when no other condition is true. Every decision needs one.
Error	Routes to error handler when a system step throws an exception.

PE Routing Expression Examples

```
// Approval decision
F_ApprovalStatus == "Approved"
F_ApprovalStatus == "Rejected"

// Rejection loop control
F_RejectionCount < 3   // Can retry
F_RejectionCount >= 3  // Max reached

// Priority routing
Strings.upper(F_Priority) == "HIGH"

// Date deadline overdue
Dates.diff(F_Deadline,
  Dates.currentDate()) < 0

// Compound logic
F_Priority == "High" &&
F_ApprovalStatus == "Pending"
```

Workflow Components

Activities

Logical groupings of steps into workflow stages. Example: SUBMISSION > REVIEW > APPROVAL > COMPLETION. Each activity can carry a milestone for SLA tracking.

Sub-processes

A Map Step launches a child workflow. Synchronous (parent waits) or asynchronous (parent continues). Used for Legal Review, multi-level approval chains, reusable logic.

Escalations

Automatic actions when a step deadline passes: reassign to supervisor, move to escalation queue, send email reminder, or abort the instance. Configured per Queue Step.

Timers

Time-based waiting in System Steps. Dates.add(Dates.currentDate(), 24, 'h') waits 24 hours. Used for cooling-off periods, retry logic, batch processing windows.

03

Build DocumentApprovalWorkflow

4 Stages · 11 Steps · Rejection Loop · Escalation · Deploy

DocumentApprovalWorkflow – Complete Map Design

Workflow Flow (4 Stages)

STAGE 1 – SUBMISSION

`LaunchStep > InitializeFields > SubmissionNotify`

STAGE 2 – REVIEW

`TrainingReviewQueue > ReviewDecision [Pass/Fail loop]`

STAGE 3 – APPROVAL

`TrainingApprovalQueue > ApprovalDecision [Approve/Reject loop]`

STAGE 4 – COMPLETION

`ApprovalNotify > MilestoneStep > TerminateStep`

Workflow Data Fields

F_Subject	String	*(empty) — Required at launch*
F_BoundDocument	String	*(empty) — Set by Process Router*
F_Priority	String	*Medium — High/Medium/Low*
F_ReviewerComments	String	*(empty) — Pass/Fail*
F_ApprovalStatus	String	*Pending — Approved/Rejected*
F_ApprovedBy	String	*(empty) — Approver username*
F_RejectionCount	Integer	*0 — Rejection loop counter*
F_ReviewCount	Integer	*0 — Review loop counter*

Escalations, Deadlines & Timers

Queue Step Escalation Configuration

```
Queue Step: TrainingApprovalQueue

// Deadline: 1 business day from entry
Dates.add(Dates.currentDate(), 1, 'd')

// Escalation: 2 hours after deadline
Escalation Type: Reassign
Reassign To: ceadmin
Delay: 2 hours

// Pre-deadline reminder (4 hours before)
Reminder Type: Email
Recipient: Current queue participant
Subject: 'Action Required: Approval Overdue'

// Escalation types available:
// - Reassign (to user or group)
// - Move to Queue (escalation queue)
// - Send Email only
// - Abort (terminate workflow)
```

Timer Expressions

2 days from now

```
Dates.add(Dates.currentDate(), 2, 'd')
```

4 hours from now

```
Dates.add(Dates.currentDate(), 4, 'h')
```

Next Monday 9AM

```
Dates.nextDay(Dates.currentDate(), 2, 9, 0)
```

1 day before deadline

```
Dates.add(F_Deadline, -1, 'd')
```

2 business days

```
Dates.addBusinessDays(Dates.currentDate(), 2)
```

04

Deploy, Test & Verify

Map > Validate · Map > Deploy to PE · Test via ACCE · Queue Stepping

Day 7 Lab Exercises

LAB-07-01 · 40 min
Install & Connect

- Verify PE service running, port 32768
- Install Process Designer V5.5.8 via IBM IM
- Configure TrainingPEConnection to localhost:32768
- Test connection — verify TrainingRoster visible
- Explore Process Designer interface panels

LAB-07-02 · 90 min
Build Workflow

- Create new map: DocumentApprovalWorkflow
- Define 8 data fields (F_Subject, F_ApprovalStatus, etc.)
- Configure LaunchStep with required fields
- Add Review Queue (TrainingReviewQueue) + escalation
- Add Approval Queue + routing decision + rejection loop

LAB-07-03 · 40 min
Deploy & Test

- Validate map — fix all ERROR items
- Deploy .pep to PE via Map > Deploy to PE
- Verify in ACCE Process Config Console
- Create test document in TrainingOS
- Step through review and approval queues end-to-end

Day 7 Complete

Workflow Development using Process Designer

Process Designer

Interface, connections, step types, data fields, map creation

Workflow Design

Routing logic, conditional branches, escalations, timers

Deploy & Test

Validate, deploy .pep, queue stepping, end-to-end verification

Next: Day 8 – IBM Content Navigator (ICN)

IBM FileNet P8 V5.5.8 | Rajesh Pasham